

How-To Guide: Reliability Engineer

Role: Reliability Engineer **Modules:** Hierarchy, Criticality, FMEA/RCM, Strategy, FMECA, Vendor Data **Pages:** Page 1 (Hierarchy), Page 2 (Criticality), Page 3 (FMEA), Page 4 (Strategy), Page 16 (FMECA)

Role Description

Reliability Engineers are responsible for building and maintaining the equipment hierarchy, assessing equipment criticality, defining failure modes, running RCM (Reliability-Centered Maintenance) decision logic, reviewing FMECA analyses, and defining the resulting maintenance tasks and work packages. They form the analytical foundation of the maintenance strategy.

Prerequisites

Requirement	Details
API server running	<code>uvicorn api.main:app --reload</code> on <code>http://localhost:8000</code>
Streamlit app running	<code>streamlit run streamlit_app/Home.py</code> on <code>http://localhost:8501</code>
Database seeded	Run seed from Page 7 for initial plant and hierarchy data
Domain knowledge	Understanding of RCM methodology, FMEA/FMECA, and OCP equipment types
Equipment catalog	Access to vendor/OEM documentation for hierarchy building

User Story RE-01: Set Up Equipment Hierarchy

Goal: Browse the existing plant hierarchy, view equipment nodes, and explore subtrees.

Step-by-Step Instructions

- 1. Navigate to Page 1 -- Hierarchy** - Open the Streamlit sidebar and click **Hierarchy**.
- 2. Select a Plant** - Use the **Select Plant** dropdown to choose a plant (e.g., `OCP-JFC1`). - If no plants exist, seed the database first (Page 7).

- 3. **View Hierarchy Statistics** - Metric cards display node counts by type. - A pie chart shows the distribution of node types (PLANT, AREA, SYSTEM, EQUIPMENT, SUB_ASSEMBLY, MAINTAINABLE_ITEM).
- 4. **Browse Hierarchy Nodes** - Use the **Filter by Type** dropdown to filter nodes:
 - ALL , PLANT , AREA , SYSTEM , EQUIPMENT , SUB_ASSEMBLY , MAINTAINABLE_ITEM
 - The data table displays: Node ID, Node Type, Name, Code, Level, Criticality, Tag.
- 5. **View Node Subtree** - Select a node from the **Select Node for Details** dropdown. - Click **Show Subtree** to display all child nodes in a hierarchical table.

API Endpoints Called

```
GET /api/v1/hierarchy/plants
GET /api/v1/hierarchy/stats?plant_id=OCP-JFC1
GET /api/v1/hierarchy/nodes?plant_id=OCP-JFC1&node_type=EQUIPMENT
GET /api/v1/hierarchy/nodes/{node_id}/tree
```

Expected Output

- A complete hierarchy tree from Plant down to Maintainable Items.
- Node statistics showing count per type.
- Subtree display for any selected node.

Error Handling

Issue	Solution
"Cannot connect" warning	Start the API server.
"No plants found"	Seed the database from Page 7 (Overview).
Empty node list	Ensure the selected plant has hierarchy data. Try filtering by a different type.

User Story RE-02: Run Criticality Assessment

Goal: Assess the criticality of equipment using a multi-criteria risk matrix (GFSN methodology) to determine risk class (I through IV).

Step-by-Step Instructions

- 1. **Navigate to Page 2 -- Criticality** - Open the Streamlit sidebar and click **Criticality**.
- 2. **Select Equipment** - The page automatically loads all EQUIPMENT type nodes. - Use the **Select Equipment** dropdown to choose an equipment item (displayed as Name (Code)).

3. **Review Current Assessment (left column)** - If an assessment already exists:

- **Risk Class:** Current risk classification (e.g., `III_HIGH`)
- **Overall Score:** Numeric score (e.g., `7.5`)
- **Status:** `DRAFT` or `APPROVED`
- For `DRAFT` assessments, click **Approve Assessment** to finalize.

4. **Run New Assessment (right column)** - Fill in the **Criticality Matrix Form**:

- Enter scores for each criterion (Safety, Environment, Production, Quality, Maintenance Cost, etc.) on a scale defined by the GFSN matrix.
- Set the **Probability** factor.
- Click **Run Assessment**.

5. **Review Results** - Success message shows: "Risk Class: {class}, Score: {score}" - Any warnings are displayed (e.g., "High safety consequence detected").

API Endpoints Called

```
GET /api/v1/hierarchy/nodes?node_type=EQUIPMENT
GET /api/v1/criticality/{node_id}
POST /api/v1/criticality/assess
{
  "node_id": "uuid-...",
  "criteria_scores": { "safety": 4, "environment": 3, "production": 5, ... },
  "probability": 3
}
PUT /api/v1/criticality/{assessment_id}/approve
```

Expected Output

```
{
  "assessment_id": "uuid-...",
  "node_id": "uuid-...",
  "overall_score": 7.5,
  "risk_class": "III_HIGH",
  "criteria_scores": { "safety": 4, "environment": 3, "production": 5 },
  "probability": 3,
  "status": "DRAFT"
}
```

Error Handling

Issue	Solution
"No equipment found"	Seed the database or create equipment nodes in the hierarchy first.
"No assessment found"	No prior assessment exists. Create one using the right-column form.
Warnings after assessment	Review high-scoring criteria. Consider additional mitigation measures.

User Story RE-03: Define Failure Modes

Goal: View existing failure modes and validate mechanism-cause combinations using the GFSN-compliant 72 valid combinations.

Step-by-Step Instructions

1. **Navigate to Page 3 -- FMEA** - Open the Streamlit sidebar and click **FMEA**. - The page has three tabs: **Failure Modes**, **Validation**, and **RCM Decision**.
2. **View Registered Failure Modes (Failure Modes Tab)** - The data table shows all failure modes with columns:
 - Failure Mode ID, What, Mechanism, Cause, Strategy Type
 - Browse to understand existing failure mode coverage.
3. **Validate a Mechanism-Cause Combination (Validation Tab)** - Switch to the **Validation** tab. - Select a **Mechanism** from the dropdown (e.g., **FATIGUE**, **CORROSION**, **EROSION**, **OVERLOAD**, **CONTAMINATION**, **DEGRADATION**). - Select a valid **Cause** from the dynamically-loaded list (the system fetches valid causes for the selected mechanism). - Click **Validate** to confirm the combination is valid. - The result shows green (valid) or red (invalid) status.
4. **View All Valid Combinations** - Below the validation form, the total number of valid combinations is shown (typically 72). - Available mechanisms are listed.

API Endpoints Called

```
GET /api/v1/fmea/failure-modes
GET /api/v1/fmea/fm-combinations?mechanism=FATIGUE
POST /api/v1/fmea/validate-combination
{ "mechanism": "FATIGUE", "cause": "CYCLIC_LOADING" }
```

Expected Output

Validation result:

```
{
  "valid": true,
  "mechanism": "FATIGUE",
  "cause": "CYCLIC_LOADING"
}
```

Error Handling

Issue	Solution
"Cannot connect"	Ensure API is running.
"Invalid combination"	The selected mechanism-cause pair is not in the GFSN-compliant set. Choose a different cause.
Empty cause dropdown	Select a mechanism first. The causes are dynamically loaded per mechanism.

User Story RE-04: Run RCM Decision Tree

Goal: Apply the RCM (Reliability-Centered Maintenance) decision logic to determine the optimal maintenance strategy for a failure mode.

Step-by-Step Instructions

- Navigate to Page 3 -- FMEA, RCM Decision Tab** - Click the **RCM Decision** tab on the FMEA page.
- Set Decision Tree Inputs (left column) - Hidden Failure:** Check if the failure is hidden (not evident to operators under normal conditions). - **Failure Consequence:** Select the consequence category:
 - `HIDDEN_SAFETY` , `HIDDEN_NONSAFETY`
 - `EVIDENT_SAFETY` , `EVIDENT_ENVIRONMENTAL`
 - `EVIDENT_OPERATIONAL` , `EVIDENT_NONOPERATIONAL`
 - Failure Pattern:** Select the degradation pattern:
 - `A_BATHTUB` , `B_AGE` , `C_FATIGUE` , `D_STRESS` , `E_RANDOM` , `F_EARLY_LIFE`
- Set Feasibility Inputs (right column) - CBM Technically Feasible:** Check if condition-based monitoring is technically possible. - **CBM Economically Viable:** Check if CBM is cost-effective. - **FT Feasible:** Check if a failure-finding task is feasible.
- Run Decision** - Click **Run RCM Decision**. - Review the results:
 - Strategy Type:** The recommended strategy (e.g., `CBM` , `TBM` , `FBM` , `RTF` , `REDESIGN`)

- **Path:** The decision tree path taken (e.g., `H -> S -> CBM_TECH -> CBM`)
- **Reasoning:** Textual explanation of why this strategy was chosen
- If a secondary task is required, a warning is displayed.

API Endpoints Called

```
POST /api/v1/fmea/rcm-decide
{
  "is_hidden": false,
  "failure_consequence": "EVIDENT_OPERATIONAL",
  "cbm technically_feasible": true,
  "cbm_economically_viable": true,
  "ft_feasible": true,
  "failure_pattern": "B_AGE"
}
```

Expected Output

```
{
  "strategy_type": "CBM",
  "path": "EVIDENT -> OPERATIONAL -> CBM_TECH_YES -> CBM_ECON_YES -> CBM",
  "reasoning": "Condition-based maintenance is both technically feasible and economically viable for this evident operational failure.",
  "requires_secondary_task": false
}
```

Error Handling

Issue	Solution
Unexpected strategy result	Review all input parameters. Ensure consequence and pattern match the actual failure mode.
"Requires secondary task" warning	A failure-finding task or additional monitoring task is needed alongside the primary strategy.

User Story RE-05: Review FMECA Analysis

Goal: Create FMECA worksheets, add failure mode rows with Severity-Occurrence-Detection (SOD) scoring, calculate RPN values, and run Stage 4 RCM decisions.

Step-by-Step Instructions

1. **Navigate to Page 16 -- FMECA Analysis** - Open the Streamlit sidebar and click **FMECA Analysis**. - The page has three tabs: **Worksheets**, **RPN Calculator**, and **Summary**.

2. **Create a Worksheet (Worksheets Tab)** - Enter:

- **Equipment ID:** Equipment identifier (e.g., EQ-001)
- **Equipment Tag:** Physical tag (e.g., BRY-SAG-ML-001)
- **Equipment Name:** Descriptive name
- **Analyst:** Your name or identifier
- Click **Create Worksheet**.
- The worksheet is created at Stage 1 with a unique ID.

3. **Add Failure Mode Rows** - After creating the worksheet, a form appears to add rows:

- **Function Description:** What the component/system does
- **Functional Failure:** How the function can fail
- **Failure Mode:** Specific failure mechanism
- **Failure Effect:** What happens when this failure occurs
- **Failure Consequence:** Category (HIDDEN_SAFETY, EVIDENT_OPERATIONAL, etc.)
- **Severity** (1-10 slider): How severe is the failure effect
- **Occurrence** (1-10 slider): How often does this failure occur
- **Detection** (1-10 slider): How easily can it be detected before failure
- Click **Add Row**. The RPN ($S \times O \times D$) is calculated automatically.

4. **View Current Rows** - A data table shows all rows with columns:

- ID, Function, Failure Mode, S, O, D, RPN, Category, Strategy.
- RPN Categories: LOW (1-49), MEDIUM (50-99), HIGH (100-199), CRITICAL (200-1000).

5. **Advance Through Stages** - Use the **Stage Management** section to advance the worksheet through stages. - Click **Advance to Stage X** to move forward.

6. **Run Stage 4 Decisions** - Click **Run Stage 4 Decisions** to automatically apply RCM logic to all rows. - Each row receives a recommended strategy type based on its consequence and feasibility.

7. **Use the RPN Calculator (RPN Calculator Tab)** - Adjust Severity, Occurrence, and Detection sliders. - The RPN is calculated in real-time with color-coded results.

8. **Review Analysis Summary (Summary Tab)** - View overall statistics:

- Total rows, Average RPN, High/Critical count, Strategies assigned
- Charts for RPN distribution and strategy distribution.
- Top risks table showing highest-RPN failure modes.
- Recommendations for improvement areas.

API Endpoints Called

```
POST /api/v1/fmea/fmeca/worksheets
  { "equipment_id": "EQ-001", "equipment_tag": "BRY-SAG-ML-001", ... }
POST /api/v1/fmea/fmeca/rpn
  { "severity": 7, "occurrence": 5, "detection": 3 }
PUT /api/v1/fmea/fmeca/worksheets/{worksheet_id}/run-decisions
GET /api/v1/fmea/fmeca/worksheets/{worksheet_id}/summary
```

Expected Output

Summary response:

```
{
  "total_rows": 12,
  "avg_rpn": 145.3,
  "high_critical_count": 4,
  "rpn_distribution": { "LOW": 3, "MEDIUM": 5, "HIGH": 3, "CRITICAL": 1 },
  "strategy_distribution": { "CBM": 5, "TBM": 3, "RTF": 2, "REDESIGN": 2 },
  "top_risks": [...],
  "recommendations": ["Address 4 high/critical RPN items", ...]
}
```

Error Handling

Issue	Solution
"Create worksheet first" message	Create a worksheet in the Worksheets tab before viewing the Summary.
Stage advance fails	Ensure prerequisite stages are completed. Check the minimum row requirements.
No strategies after Stage 4	Ensure all rows have failure consequence values set.

User Story RE-06: Define Maintenance Tasks

Goal: View and validate maintenance tasks that result from the FMEA/RCM analysis.

Step-by-Step Instructions

1. **Navigate to Page 4 -- Strategy, Tasks Tab** - Open the Streamlit sidebar and click **Strategy**. - The **Tasks** tab is selected by default.
2. **Review Maintenance Tasks** - The data table shows all maintenance tasks:
 - Task ID, Name, Task Type, Status
 - Task types include: INSPECT, CHECK, TEST, LUBRICATE, CLEAN, REPLACE, REPAIR, CALIBRATE.

3. **Validate Task Naming** - Switch to the **Naming Validation** tab. - Enter a task name and select the task type. - Click **Validate Task Name**. - The system checks the name against GFSN naming conventions.

API Endpoints Called

```
GET /api/v1/tasks/
POST /api/v1/tasks/validate-name { "name": "INSPECT_BEARING_TEMP", "task_type": "INSPECT" }
```

Error Handling

Issue	Solution
Empty task list	Run FMEA analysis and RCM decisions first. Tasks are generated from the strategy flow.
Naming validation fails	Follow the specific issue descriptions in the warning messages to correct the name.

User Story RE-07: Build Hierarchy from Vendor Data

Goal: Automatically generate a complete equipment hierarchy (equipment, sub-assemblies, maintainable items) from vendor/OEM specifications.

Step-by-Step Instructions

1. **Navigate to Page 1 -- Hierarchy** - Scroll down to the **Build from Vendor Data** section (below the node browser).
2. **Fill in the Vendor Data Form - Area Code:** 3-4 character area identifier (e.g., **BRY**) - **Equipment Type:** Select from the dropdown:
 - SAG_MILL, BALL_MILL, ROD_MILL, SLURRY_PUMP, FLOTATION_CELL,
 - BELT_CONVEYOR, THICKENER, BELT_FILTER, ROTARY_DRYER, CRUSHER,
 - VIBRATING_SCREEN, HYDROCYCLONE, AGITATOR, COMPRESSOR, HEAT_EXCHANGER
 - **Model:** Equipment model number (e.g., **36x20**)
 - **Manufacturer:** OEM name (e.g., **FLSmidth**)
 - **Power (kW):** Rated power
 - **Weight (kg):** Equipment weight
3. **Build Hierarchy** - Click **Build Hierarchy** (the form submit button). - The AI engine generates a complete hierarchy based on the equipment type and vendor data.

4. **Review Results** - Success message shows: "Nodes Created: X" with the count of generated nodes. - Full JSON response shows the complete hierarchy structure. - Browse the newly created nodes in the hierarchy browser above.

API Endpoints Called

```
POST /api/v1/hierarchy/build-from-vendor
{
  "plant_id": "OCP-JFC1",
  "area_code": "BRY",
  "equipment_type": "SAG_MILL",
  "model": "36x20",
  "manufacturer": "FLSmidth",
  "power_kw": 12000.0,
  "weight_kg": 450000.0
}
```

Expected Output

```
{
  "nodes_persisted": 15,
  "equipment_node": { "node_id": "uuid-...", "name": "SAG Mill 36x20", "tag": "BRY-SAG-ML-001" },
  "sub_assemblies": [
    { "name": "Mill Shell", "type": "SUB_ASSEMBLY" },
    { "name": "Drive System", "type": "SUB_ASSEMBLY" },
    { "name": "Lubrication System", "type": "SUB_ASSEMBLY" }
  ],
  "maintainable_items": [
    { "name": "Drive Motor Bearings", "type": "MAINTAINABLE_ITEM" },
    { "name": "Pinion Gear", "type": "MAINTAINABLE_ITEM" }
  ]
}
```

Error Handling

Issue	Solution
"Missing required fields" (422)	Ensure plant_id, area_code, and equipment_type are all provided.
Build returns error (400)	Check that the plant exists. Verify the area code is valid.
Fewer nodes than expected	The hierarchy depth depends on the equipment type. Some types have simpler structures.

Quick Reference: Reliability Engineer Workflow

Page 1: Hierarchy

```
+-----+
| Browse hierarchy |
| Build from vendor |
+-----+
```

Page 2: Criticality

```
+-----+
| Assess criticality |
| Approve assessment |
+-----+
```

Page 3: FMEA

```
+-----+
| Define failure   |
| modes & validate |
| Run RCM decisions |
+-----+
```

|

Page 4: Strategy

```
+-----+
| Review tasks     |
| Validate names   |
| Approve work pkgs |
+-----+
```

Page 16: FMECA

```
+-----+
| Create worksheets |
| Add rows (SOD)   |
| Run Stage 4      |
| Review summary   |
+-----+
```